

Address Translation

At the end of this episode, I will be able to:

1. Address translation

Learner Objective: *Learn about address translation in modern networks including network address translation and port address translation*

Description: In this episode, you will learn about address translation in modern networks. This includes a discussion of NAT and PAT. This episode also walks you through a sample configuration on a Cisco router.

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- When it was clear there would be a shortage of IPv4 addresses with the explosion of the public Internet, the standards bodies created RFC 1918 Private Addresses
 - These addresses may be used internally on networks and then translated to permit communication on the public Internet
 - The RFC 1918 addresses are - 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16
 - Network Address Translation (NAT) can be used to make a one to one translation of a private use only address (for example 10.10.10.100) to an Internet routable public address (for example, 67.10.23.78)
 - Port Address Translation (PAT) allows a single public IP address to be shared by multiple internal private use only clients - arbitrary "port" numbers are used in order to track the internal clients using the single public address
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Additional Resources:

Network Address Translation

https://en.wikipedia.org/wiki/Network_address_translation